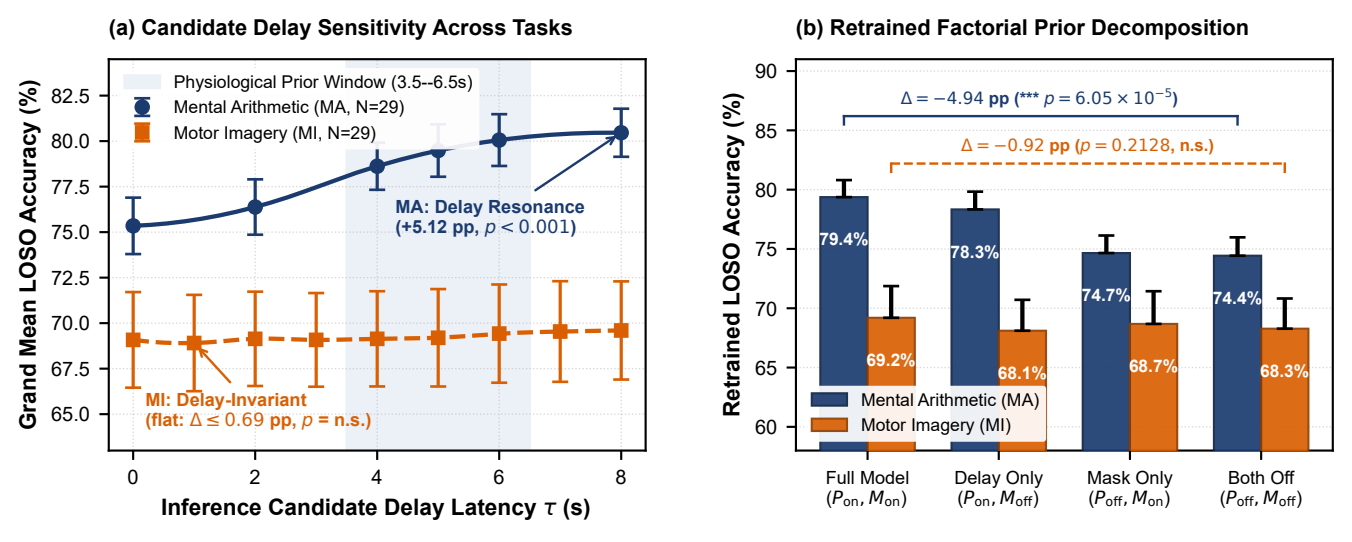

(a) Candidate Delay Sensitivity Across Tasks
Physiological Prior Window (3.5--6.5s)
Mental Arithmetic (MA, N=29)
Motor Imagery (MI, N=29)
MA: Delay Resonance
(+5.12 pp, p < 0.001)
MI: Delay-Invariant
(flat: Δ ≤ 0.69 pp, p = n.s.)
Grand Mean LOSO Accuracy (%)
Inference Candidate Delay Latency τ (s)
(b) Retrained Factorial Prior Decomposition
Δ = −4.94 pp (*** p = 6.05 × 10⁻⁵)
Δ = −0.92 pp (p = 0.2128, n.s.)
79.4%
69.2%
78.3%
68.1%
74.7%
68.7%
74.4%
68.3%
Mental Arithmetic (MA)
Motor Imagery (MI)
Retrained LOSO Accuracy (%)
Full Model
(Pon, Mon)
Delay Only
(Pon, Moff)
Mask Only
(Poff, Mon)
Both Off
(Poff, Moff)